1. What is the main purpose of using maps in data visualization in business intelligence dashboards?

Maps help visualize **geographical patterns and trends** in business data.

Example: A sales dashboard can use a map to show **sales revenue by city**, helping managers identify regions with high or low sales.

2. Explain the difference between Geocoding and Reverse Geocoding.

- **Geocoding:** Converts an address or place name into **Latitude and Longitude**.
- **Reverse Geocoding:** Converts **Latitude and Longitude** into a location name or address.

In Power BI, geocoding helps map visuals locate cities, addresses, or coordinates correctly.

3. Why are Latitude and Longitude more accurate than City names?

Latitude and Longitude identify an **exact geographical location**, while city names can be ambiguous.

Example: If a company has stores in different cities with similar names, coordinates ensure that each store appears at the correct location on the map.

4. What problems can occur when using only City and Country fields?

Two common problems are:

1. **Ambiguous city names** — the same city name can exist in different countries or regions.
2. **Incorrect location detection** — Power BI may place a city in the wrong geographical location.

5. What is the role of formatting in map visuals?

Formatting makes geographical data easier to understand:

- **Colors** show categories or performance.
- **Size** shows the magnitude of values.
- **Tooltips** provide additional details.
- **Legends** explain colors and categories.

This helps users quickly identify important business areas and make better decisions.

6. What are annotations in dashboard maps?

Annotations are **labels, notes, or markers** added to a map to highlight important information.

They are useful for storytelling because they explain **why a specific location or region is important**, such as highlighting a region with unusually high sales or a new store opening.